

LAB MANUAL



**Faculty of Computing, Engineering and The
Built Environment**

Week 2 - Azure SQL

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1. What is Azure SQL?

Azure SQL is a family of managed, secure, and intelligent products that use the SQL Server database engine in the Azure cloud.

- **Azure SQL Database:** Support modern cloud applications on an intelligent, managed database service, that includes serverless compute.
- **Azure SQL Managed Instance:** Modernize your existing SQL Server applications at scale with an intelligent fully managed instance as a service, with almost 100% feature parity with the SQL Server database engine. Best for most migrations to the cloud.
- **SQL Server on Azure VMs:** Lift-and-shift your SQL Server workloads with ease and maintain 100% SQL Server compatibility and operating system-level access.

Azure SQL is built upon the familiar SQL Server engine, so you can migrate applications with ease and continue to use the tools, languages, and resources you're familiar with. Your skills and experience transfer to the cloud, so you can do even more with what you already have.

2. Overview

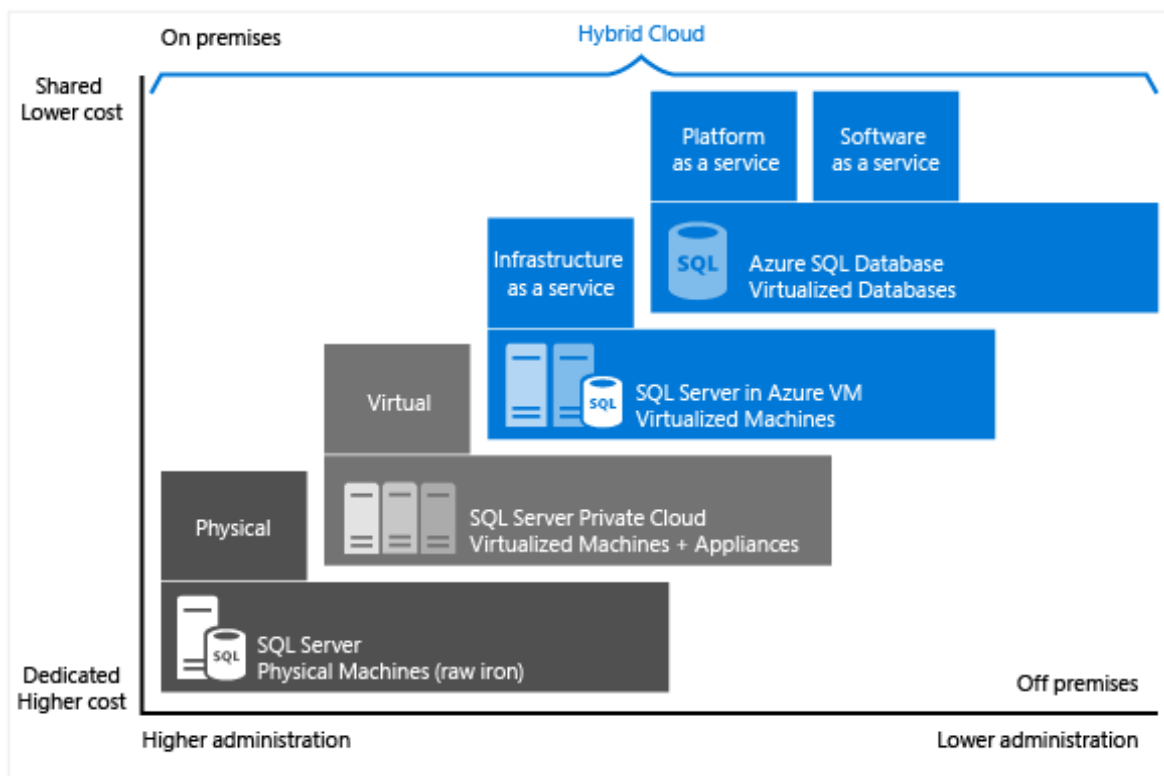
In today's data-driven world, driving digital transformation increasingly depends on our ability to manage massive amounts of data and harness its potential. But today's data estates are increasingly complex, with data hosted on-premises, in the cloud, or at the edge of the network. Developers who are building intelligent and immersive applications can find themselves constrained by limitations that can ultimately impact their experience. Limitations arising from incompatible platforms, inadequate data security, insufficient resources and price-performance barriers create complexity that can inhibit app modernization and development.

One of the first things to understand in any discussion of Azure versus on-premises SQL Server databases is that you can use it all. Microsoft's data platform leverages SQL Server technology and makes it available across physical on-premises machines, private cloud environments, third-party hosted private cloud environments, and the public cloud.

3. Business motivations

There are several factors that can influence your decision to choose between the different data offerings:

- (1) **Cost:** Both PaaS and IaaS option include base price that covers underlying infrastructure and licensing. However, with IaaS option you need to invest additional time and resources to manage your Database, while in PaaS you get these administration features included in the price. **IaaS enables you to shut down resources while you are not using them to decrease the cost**, while PaaS is always running unless you drop and re-create your resources when they are needed.
- (2) **Administration:** PaaS options reduce the amount of time that you need to invest to administer the Database. However, it also limits the range of custom administration tasks and scripts that you can perform or run. For example, the CLR is not supported with SQL Database, but is supported for an instance of SQL Managed Instance. Also, no deployment options in PaaS support the use of trace flags.
- (3) **Service-level agreement:** Both IaaS and PaaS provide high, industry standard SLA. PaaS option guarantees 99.99% SLA, while IaaS guarantees 99.95% SLA for infrastructure, meaning that you need to implement additional mechanisms to ensure availability of your databases. You can attain 99.99% SLA by creating an additional SQL virtual machine, and implementing the SQL Server Always On availability group high availability solution.
- (4) **Time to move to Azure:** SQL Server on Azure VM is the exact match of your environment, so migration from on-premises to the Azure VM is no different than moving the databases from one on-premises Server to another. SQL Managed Instance also enables easy migration; however, there might be some changes that you need to apply before your migration.



As seen in the above figure, each service offering can be characterized by the level of administration you have over the infrastructure, and by the degree of cost efficiency.

In Azure, you can have your SQL Server workloads running as a hosted service (PaaS), or a hosted infrastructure (IaaS). Within PaaS, you have multiple product options, and service tiers within each option.

4. Azure SQL Database

Azure SQL Database is a relational database-as-a-service (DBaaS) hosted in Azure that falls into the industry category of Platform-as-a-Service (PaaS).

Best for modern cloud applications that want to use the latest stable SQL Server features and have time constraints in development and marketing.

A fully managed SQL Server database engine, based on the latest stable Enterprise Edition of SQL Server. SQL Database has two deployment options built on standardized hardware and software that is owned, hosted, and maintained by Microsoft.

With SQL Server, you can use built-in features and functionality that requires extensive configuration (either on-premises or in an Azure virtual machine). When using SQL Database, you pay-as-you-go with options to scale up or out for greater power with no interruption. SQL Database has some additional features that are not available in SQL Server, such as built-in high availability, intelligence, and management.

Azure SQL Database offers the following deployment options:

As a [single database](#) with its own set of resources managed via a [logical SQL server](#). A single database is similar to a [contained database](#) in SQL Server. This option is optimized for modern application development of new cloud-born applications. [Hyperscale](#) and [serverless](#) options are available.

An [elastic pool](#), which is a collection of databases with a shared set of resources managed via a logical SQL server. Single databases can be moved into and out of an elastic pool. This option is optimized for modern application development of new cloud-born applications using the multi-tenant SaaS application pattern. Elastic pools provide a cost-effective solution for managing the performance of multiple databases that have variable usage patterns.

5. Azure SQL Managed Instances?

[Azure SQL Managed Instance](#) falls into the industry category of *Platform-as-a-Service (PaaS)*, and is best for most migrations to the cloud. SQL Managed Instance is a collection of system and user databases with a shared set of resources that is lift-and-shift ready.

- 1) Best for new applications or existing on-premises applications that want to use the latest stable SQL Server features and that are migrated to the cloud with minimal changes. An instance of SQL Managed Instance is similar to an instance of the [Microsoft SQL Server database engine](#) offering shared resources for databases and additional instance-scoped features.
- 2) SQL Managed Instance supports database migration from on-premises with minimal to no database change. This option provides all of the PaaS benefits of Azure SQL Database but adds capabilities that were previously only available in SQL Server VMs. This includes a native virtual network and near 100% compatibility with on-premises SQL Server. Instances of SQL Managed Instance provide full SQL Server access and feature compatibility for migrating SQL Servers to Azure.

6. SQL Server on Azure VM

[SQL Server on Azure VM](#) falls into the industry category *Infrastructure-as-a-Service (IaaS)* and allows you to run SQL Server inside a fully managed virtual machine (VM) in Azure.

1. SQL Server installed and hosted in the cloud runs on Windows Server or Linux virtual machines running on Azure, also known as an infrastructure as a service (IaaS). SQL virtual machines are a good option for migrating on-premises SQL Server databases and applications without any database change. All recent versions and editions of SQL Server are available for installation in an IaaS virtual machine.
2. **Best for migrations and applications requiring OS-level access.** SQL virtual machines in Azure are lift-and-shift ready for existing applications that require fast migration to the cloud with minimal changes or no changes. SQL virtual machines offer full administrative control over the SQL Server instance and underlying OS for migration to Azure.
3. The most significant difference from SQL Database and SQL Managed Instance is **that SQL Server on Azure Virtual Machines allows full control over the database engine.** You can choose when to start maintenance/patching, change the recovery model to simple or bulk-logged, pause or start the service when needed, and you can fully customize the SQL Server database engine. With this additional control comes the added responsibility to manage the virtual machine.
4. Rapid development and test scenarios when you do not want to buy on-premises non-production SQL Server hardware. SQL virtual machines also run on standardized hardware that is owned, hosted, and maintained by Microsoft. When using SQL virtual machines, you can either pay-as-you-go for a SQL

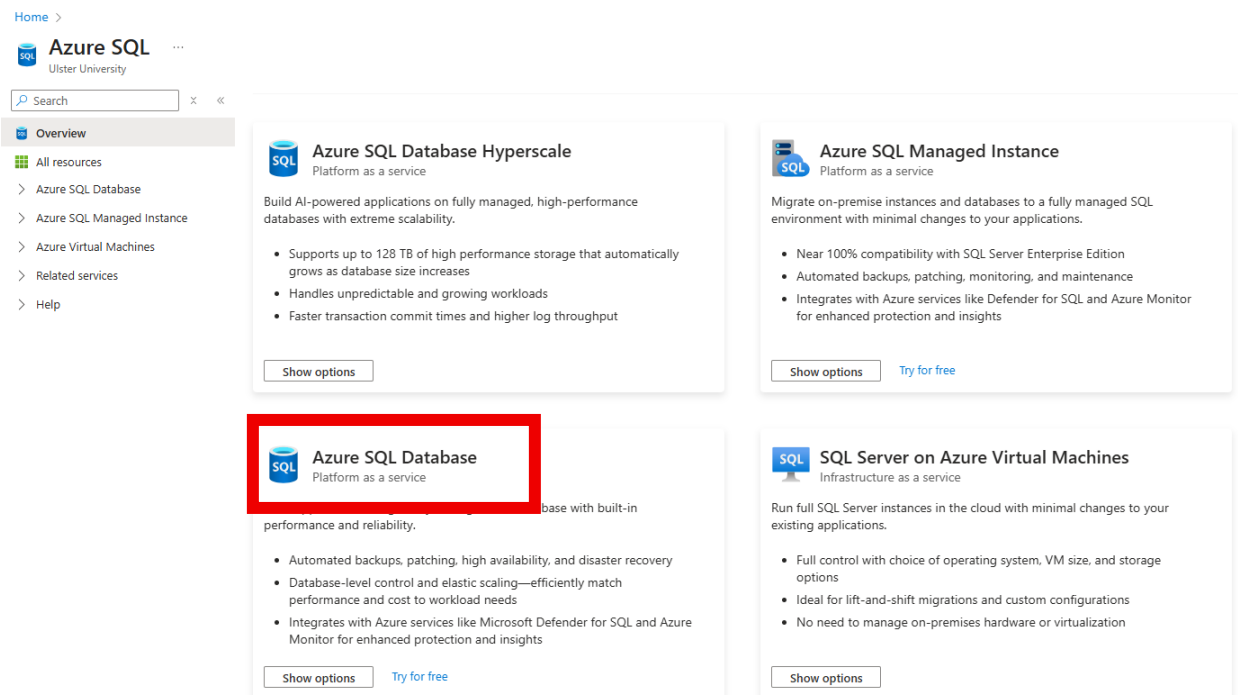
Server license already included in a SQL Server image or easily use an existing license. You can also stop or resume the VM as needed.

5. Optimized for migrating existing applications to Azure or extending existing on-premises applications to the cloud in hybrid deployments. In addition, you can use SQL Server in a virtual machine to develop and test traditional SQL Server applications. With SQL virtual machines, you have the full administrative rights over a dedicated SQL Server instance and a cloud-based VM. It is a perfect choice when an organization already has IT resources available to maintain the virtual machines. These capabilities allow you to build a highly customized system to address your application's specific performance and availability requirements.

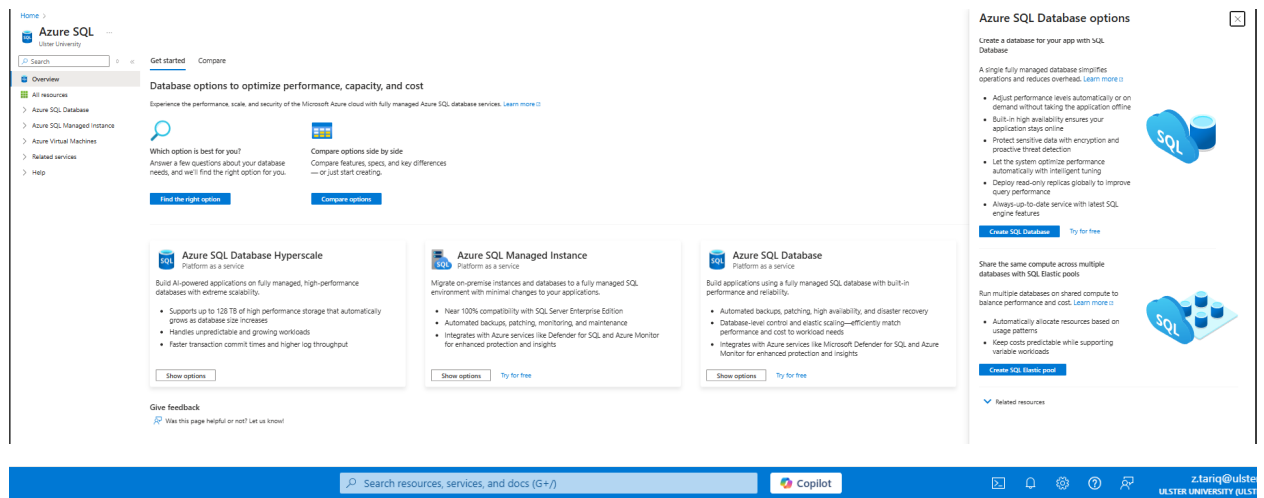
7. Create and manage Azure SQL resources with the Azure portal

To create an Azure SQL Database, go to the following link.

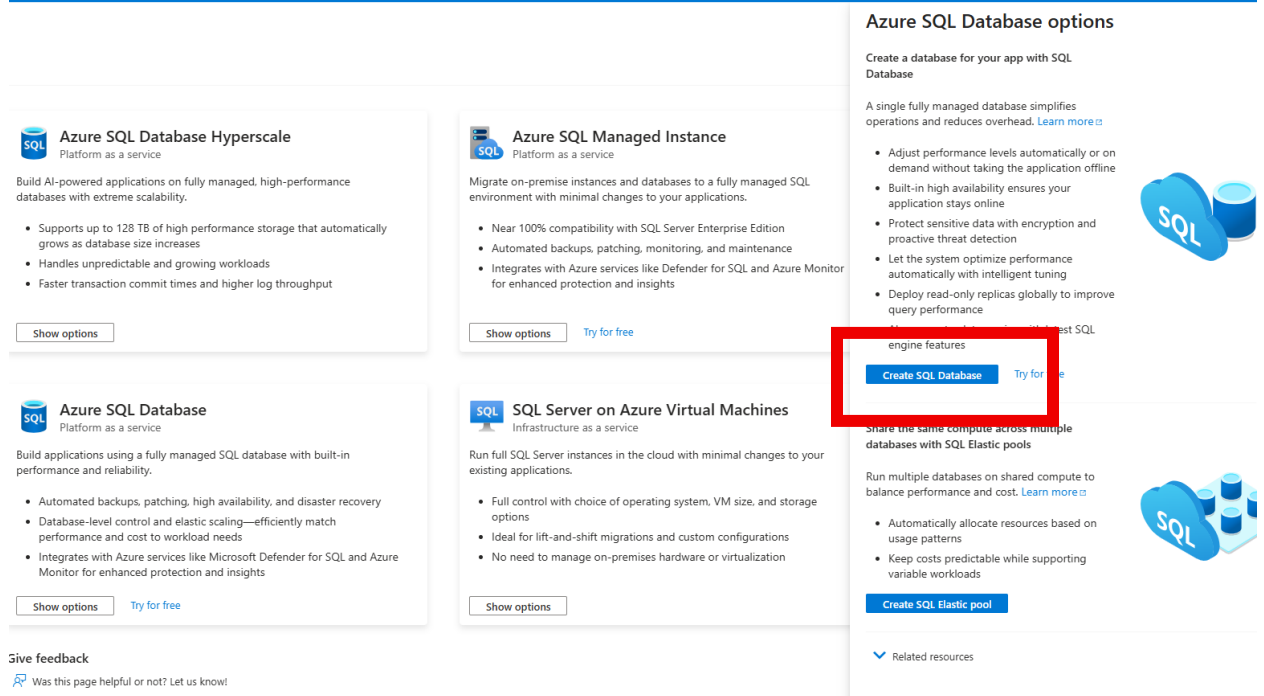
<https://portal.azure.com/#create/Microsoft.AzureSQL> (you should be signed in to your Azure account).



1. Click on **Azure SQL databases**, click on show options, **database**, and click on select **Create SQL Database**.



The screenshot shows the 'Azure SQL Database options' page. On the left, there's a navigation pane with 'Azure SQL Database' selected. The main content area displays several options: 'Azure SQL Database Hyperscale', 'Azure SQL Managed Instance', and 'Azure SQL Database'. Each option has a 'Show options' button. A red box highlights the 'Create SQL Database' button in the 'Azure SQL Database' section. On the right, there's a sidebar with 'Azure SQL Database options' and a 'Create SQL Database' button.



The screenshot shows the 'Create SQL Database' form. The 'Basics' tab is selected. The 'Subscription' dropdown is set to 'Azure for Students'. The 'Resource group' dropdown is set to 'Create new', and the 'Database name' field contains 'mySampleDatabase'. The 'Create SQL Database' button is highlighted with a red box.

2. On the **Basics** tab of the **Create SQL Database** form, under **Subscription**, select the 'Azure for Students'
3. For **Resource group**, select **Create new**, enter *myResourceGroup*, and select **OK**.
4. For **Database name**, enter *mySampleDatabase*.

Create SQL Database

Microsoft

[✕ Basics](#)
[Networking](#)
[Security](#)
[Additional settings](#)
[Tags](#)
[Review + create](#)

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Did you know that new users in Azure can create a free Azure SQL Database and use it for 12 months using Azure free account? [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * [?](#)

Resource group * [?](#) [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server * [?](#) [Create new](#)

✓ Database name should
 ✓ The value has a length
 ✓ Database name should
 ✓ No database with the s

✕ The value must not be empty.

Want to use SQL elastic pool? [?](#) ☐ Yes ☒ No

5. For **Server**, select **Create new**, and fill out the **New server** form with the following values:
 - 5.1. **Server name:** Enter *mysqlserver*, and add some characters for uniqueness. We can't provide an exact server name to use because server names must be globally unique for all servers in Azure, not just unique within a subscription. So enter something like *mysqlserver12345*, and the portal lets you know if it's available or not.
 - 5.2. **Location:** Select a location **Europe() UK South**), from the dropdown list.
 - 5.3. **Authentication:** Use 'Use SQL authentication'
 - 5.4. **Server admin login:** Enter *azureuser*.
 - 5.5. **Password:** Enter a password that meets requirements and enter it again in the **Confirm password** field. For example 'ulster@123'
 - 5.6. Select **OK**.

[Home](#) > [Azure SQL](#) > [Create SQL Database](#) >

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * .database.windows.net

Location * (Europe) UK South 

Authentication

 Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) 

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#)  using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) , or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login *

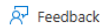
Password * 

Confirm password * 

OK

6. Leave **Want to use SQL elastic pool** set to **No**.
7. Under **Compute + storage**, select **Configure database**.
8. This quickstart uses a serverless database, so select **Serverless**, and then select **Apply**.
9. Leave defaults for Max vCores, Min vCores, Auto-pause delay.
10. **Backup Storage redundancy** : Select Locally-redundant backup storage

Configure ...



Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

 SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier General Purpose (Most budget friendly) ▾
[Compare service tiers](#)

Compute tier

☐ **Provisioned** - Compute resources are pre-allocated. Billed per hour based on vCores configured.

☒ **Serverless** - Compute resources are auto-scaled. Billed per second based on vCores used.

Compute Hardware

Select the hardware configuration based on your workload requirements. Availability of compute optimized, memory optimized, and confidential computing hardware depends on the region, service tier, and compute tier.

Hardware Configuration

Standard-series (Gen5)
 up to 80 vCores, up to 240 GB memory
[Change configuration](#)

Max vCores

2

Min vCores

0.5 vCores

2.02 GB MIN MEMORY 3 GB MAX MEMORY

Apply



Cost summary

General Purpose (GP_S_Gen5_1)
 Cost per GB (in USD) 0.13
 Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.50 USD
COMPUTE COST / VCORE SECOND ¹ 0.000181 USD

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Select **Next: Networking** at the bottom of the page.

11. On the **Networking** tab, for **Connectivity method**, select **Public endpoint**.

12. For **Firewall rules**,

- 12.1. Leave **Allow Azure services and resources to access this Server** set to **No**.
- 12.2. set **Add current client IP address** to **Yes**.

13. **Connection policy**: Default

14. **Encrypted connections**: Default

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'mysqlserverzeeshan' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method * ⓘ

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

Add current client IP address *

15. Next in **Security** settings, leave everything by default.
16. Select **Next: Additional settings** in upper tab of the page.
17. On the **Additional settings** tab, in the **Data source** section, for **Use existing data**, select **Sample**. This creates an AdventureWorksLT sample database so there's some tables and data to query and experiment with, as opposed to an empty blank database. Click **OK**

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data *

None Backup **Sample**

AdventureWorksLT will be created as the sample database.

AdventureWorksLT

Selecting this sample will modify the "Compute + Storage" settings configured in Basics, for backup compatibility. Visit "Basics" or "Review + Create" tabs to review the changes.

Do you want to continue?

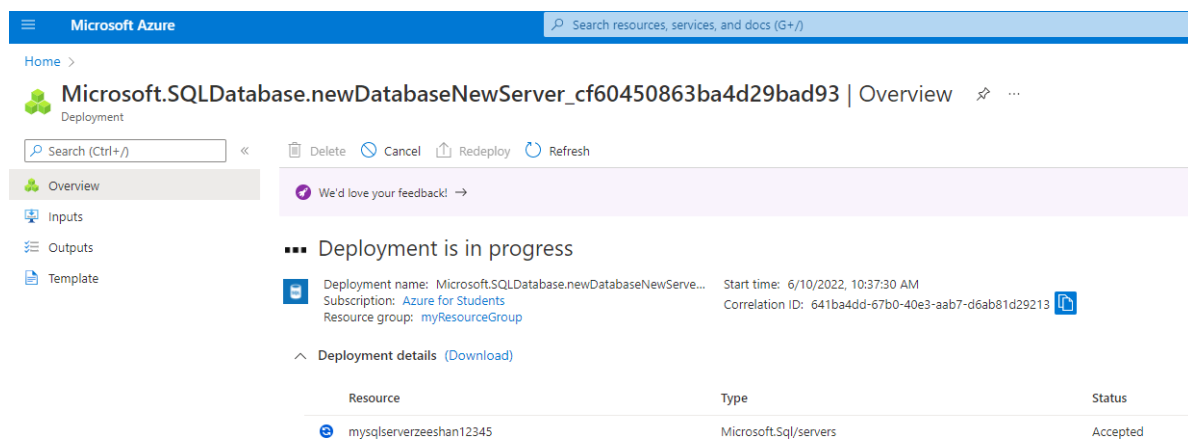
OK

Cancel

18. Select **Review + create** at the bottom of the page:

19. On the **Review + create** page, after reviewing, select **Create**.

You will see the screen like this:



The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo and a search bar. Below the navigation bar, the breadcrumb trail reads: Home > Microsoft.SqlDatabase.newDatabaseNewServer_cf60450863ba4d29bad93 | Overview. The main content area shows the deployment progress of a new SQL Database. A message at the top says "We'd love your feedback." Below that, a status bar indicates "Deployment is in progress". The deployment details show the deployment name as "Microsoft.SqlDatabase.newDatabaseNewServe...", the subscription as "Azure for Students", and the resource group as "myResourceGroup". The start time is "6/10/2022, 10:37:30 AM" and the correlation ID is "641ba4dd-67b0-40e3-aab7-d6ab81d29213". A table below shows the deployment details:

Resource	Type	Status
mysqlserverzeeshan12345	Microsoft.Sql/servers	Accepted

Deployment will take few minutes. Once deployment is complete, Go to resource

Home > Microsoft.SQLDatabase.newDatabaseNewServer_990ca865bd8e41b18bf3f | Overview ...

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_990ca865bd8e41b18bf3f

Subscription : Azure for Students

Resource group : myResourceGroup

Start time : 9/28/2025, 9:10:32 AM

Correlation ID : d9291216-fe45-414d-b0d3-bcc3209572b

> Deployment details

> Next steps

Go to resource

8. Query the Database

Once your Database is created, you can use the **Query Editor (preview)** in the Azure portal to connect to the database and query data.

Home > Microsoft.SQLDatabase.newDatabaseNewServer_990ca865bd8e41b18bf3f | Overview >

mySampleDatabase (mysqlserverkamal1234/mySampleDatabase)

SQL database

Search

Copy Restore Export Set server firewall Delete Connect with... Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Essentials

Resource group (copy)

Status

Location

Subscription (copy)

Subscription ID

Tags (add)

Server name

Connection strings

Pricing tier

Auto-pause delay

Earliest restore point

Getting started

Monitoring

Properties

Features

Notifications (0)

Integrations

Tutorials

Start working with your database

Connect to your database and start working with data with a few simple steps. [Learn more](#)

Configure access

Configure network access to your SQL server. [Learn more](#)

Configure

Connect to application

Use connection strings to connect to your SQL database from your applications and favorite tools. [See connection strings](#)

Start developing

Work in your database by using tools to add, modify and query data. [Compare tools](#)

Open Azure Data Studio

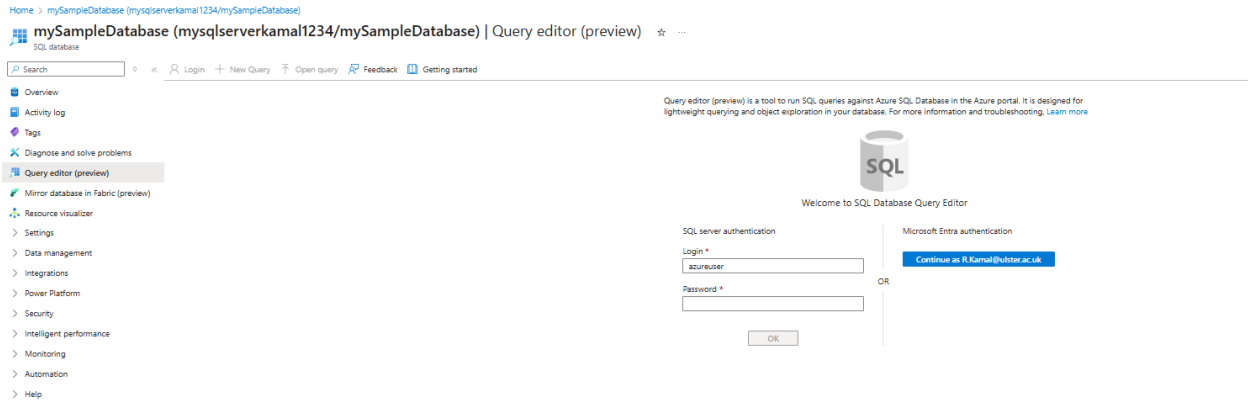
Open in Visual Studio

Open in Visual Studio Code

Mirror database in Fabric

Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

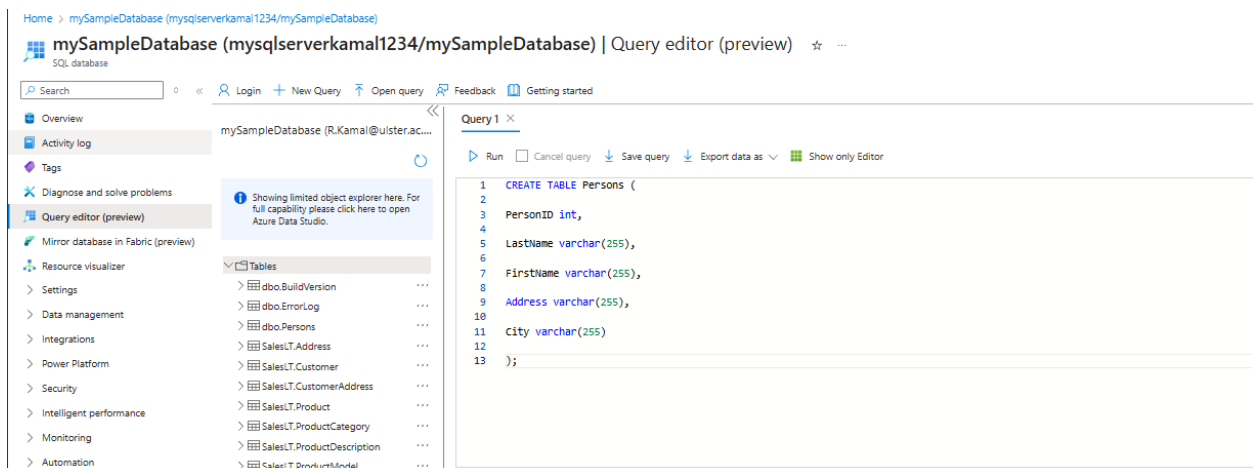
1. Enter your server admin login information, and select **OK**.
2. If it prompts to add your ip address, add your ip address and then click **OK**.



- Enter the following query in the **Query editor** pane.
- Run a **CREATE** query

SQLCopy

```
CREATE TABLE Persons (
    PersonID int,
    LastName varchar(255),
    FirstName varchar(255),
    Address varchar(255),
    City varchar(255)
);
```



- Select **Run**, and then review the query results in the **Results** pane.

6. Close the **Query editor** page, and select **OK** when prompted to discard your unsaved edits.

Go to Query editor again.

1) Run a **SELECT** query

```
SELECT TOP 20 pc.Name as CategoryName, p.name as ProductName
FROM SalesLT.ProductCategory pc
JOIN SalesLT.Product p
ON pc.productcategoryid = p.productcategoryid;
```

Query 1 ×

▶ Run
☐ Cancel query
⬇ Save query
⬇ Export data as ▾
■ Show only Editor

```

1 SELECT TOP 20 pc.Name as CategoryName, p.name as ProductName
2 FROM SalesLT.ProductCategory pc
3 JOIN SalesLT.Product p
4 ON pc.productcategoryid = p.productcategoryid;
```

Results Messages

CategoryName	ProductName
Road Frames	HL Road Frame - Black, 58
Road Frames	HL Road Frame - Red, 58
Helmets	Sport-100 Helmet, Red
Helmets	Sport-100 Helmet, Black

2) Run an **INSERT** query

To add a new product to the SalesLT.Product table, run the following INSERT T-SQL statement.

For that, click on +New Query:

```
INSERT INTO [SalesLT].[Product]
( [Name]
, [ProductNumber]
, [Color]
, [ProductCategoryID]
, [StandardCost]
, [ListPrice]
, [SellStartDate]
)
VALUES
('myNewProduct'
,123456789
,'NewColor'
,1
,100
,100
,GETDATE() );
```

Query 1 ✕

 Run ☐ Cancel query  Save query  Export data as  Show only Editor

8		, [sellstartdate]
9		)
10	VALUES	
11		('myNewProduct'
12		,123456789
13		, 'NewColor'
14		,1
15		,100
16		,100
17		,GETDATE());

Results Messages

Query succeeded: Affected rows: 1

3) Run a **DELETE** query

Run the following DELETE T-SQL statement to remove your new product.

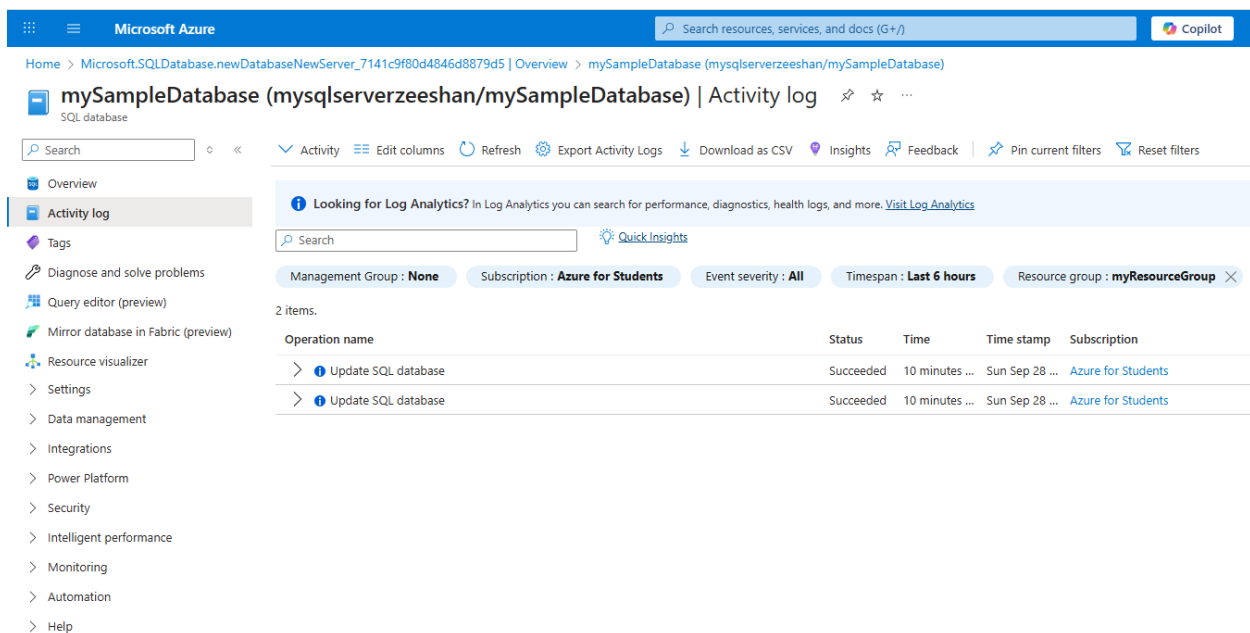
```
DELETE FROM [SalesLT].[Product]

WHERE Name = 'myNewProduct';
```

Select **Run** to delete the specified row in the Product table. The **Messages** pane displays **Query succeeded: Affected rows: 1.**

Close and Clean up resources

Click on **Activity Log** to see above activity recorded.



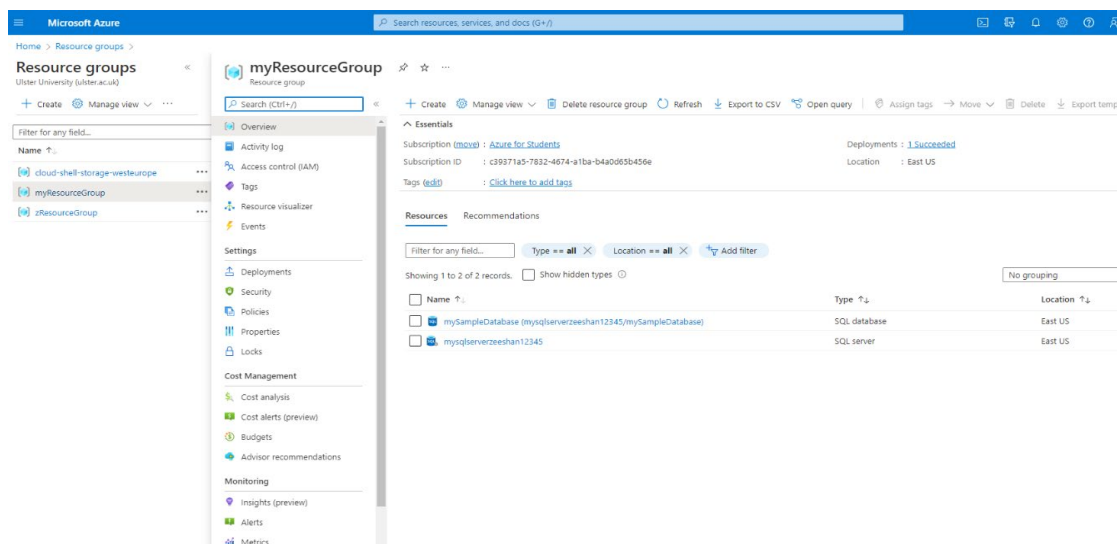
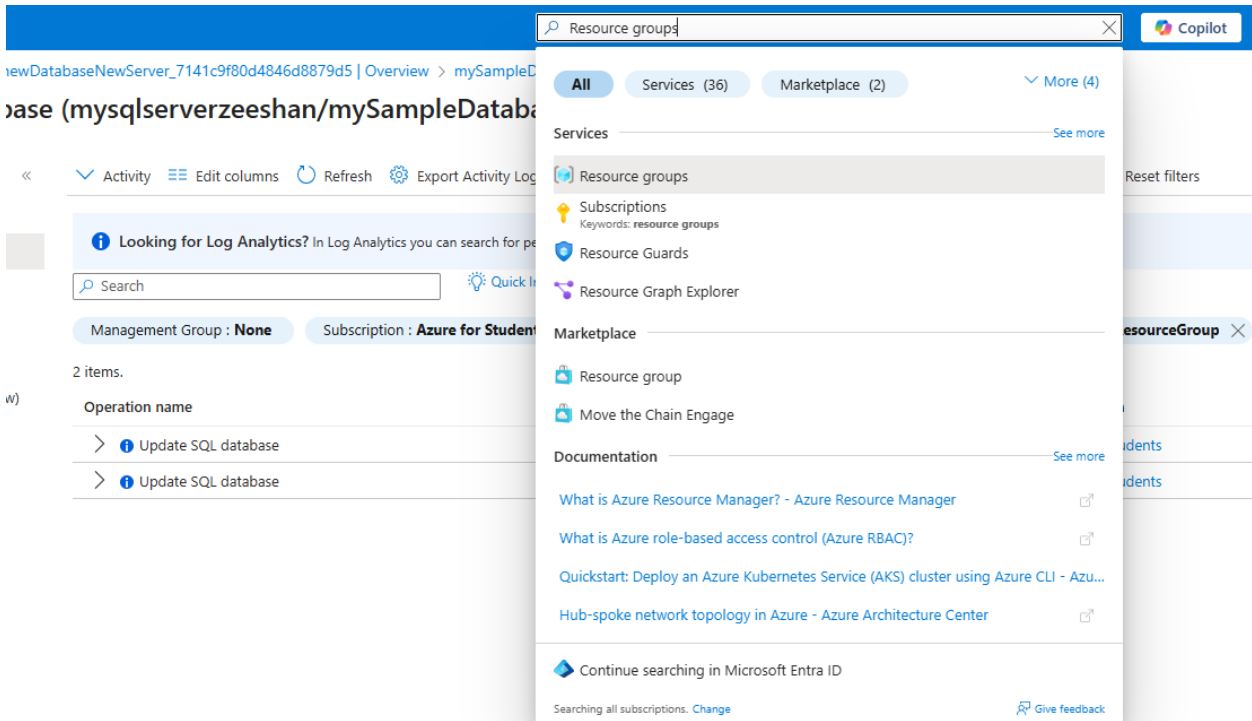
The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The breadcrumb trail indicates the path: Home > Microsoft.SQLDatabase.newDatabaseNewServer_7141c9f80d4846d8879d5 | Overview > mySampleDatabase (mysqlserverzeeshan/mySampleDatabase). The main heading is "mySampleDatabase (mysqlserverzeeshan/mySampleDatabase) | Activity log". Below this, there's a search bar and a set of filters: Management Group: None, Subscription: Azure for Students, Event severity: All, Timespan: Last 6 hours, and Resource group: myResourceGroup. The activity log shows 2 items, both of which are "Update SQL database" operations that succeeded 10 minutes ago on Sun Sep 28. The left sidebar contains a navigation menu with options like Overview, Activity log (selected), Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, Intelligent performance, Monitoring, Automation, and Help.

Keep the resource group, Server, and single Database to go on to the next steps, and learn how to connect and query your Database with different methods.

When you're finished using these resources, you can delete the resource group you created, which will also

To delete **myResourceGroup** and all its resources using the Azure portal:

1. In the portal, search for and select **Resource groups**, and then select **myResourceGroup** from the list.



2. On the resource group page, select your resource group
3. and click on **Delete resource group**.

myResourceGroup
Resource group

How to manage these changes more efficiently with deployment tools? [Generate Bicep code to](#)

Search [+ Create](#) [Manage view](#) [Delete resource group](#) [Refresh](#) [Export to CSV](#) [Open qu](#)

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Subscription ([move](#)) : [Azure for Students](#) Di

Subscription ID : 4fc85f45-4c0a-4fa2-872e-d0e5975f54b6 Lc

Tags ([edit](#)) : [Add tags](#)

Resources Recommendations

[Filter for any field...](#) [Type equals all](#) [Location equals all](#) [+ Add filter](#)

☐	Name ↑	Type
☐	mySampleDatabase (mysqlserverzeeshan/mySampleDatabase) ...	SQL
☐	mysqlserverzeeshan ...	SQL

4. Under **Type** the resource group name, enter *myResourceGroup*, and then select **Delete**.

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

myResourceGroup

Dependent resources to be deleted (3)
All dependent resources, including hidden types, are shown

Name	Resource type
master (mysqlserverzeeshan/master)	SQL database
mySampleDatabase (mysqlserverzeeshan/mySam	SQL database
mysqlserverzeeshan	SQL server

Delete confirmation

Deleting this resource group and its dependent resources is a permanent action and cannot be undone.

[Delete](#) [Go back](#)

Lab Tasks

Scenario

A university wishes to store information about students, modules, courses and lecturers.

Courses have a unique course code, course name and an accreditation level, which is an enumeration between 4 and 8. Each course has a coordinator, which is a lecturer. Courses consist of modules; the number of modules vary depending on the number of credit points that can be earned in each module. Modules have a name, unique module code and are delivered by a nominated lecturer. Lecturers have unique employee numbers, forenames, middle names and surnames, a date of birth, an email address and a rank of seniority. The rank of seniority is an numerical representation (enumeration) between 1 and 40. Students have unique student numbers, forenames, middle names and surnames, a date of birth, an email address and a current enrollment level, an enumeration between 1 and 20.

Task 1:

I. From the above scenario, produce a list of all attributes that need to be stored.

Note the data type that each attribute will be. A list of available data types is presented in the MSSQL manual which is available online at

<https://docs.microsoft.com/en-us/sql/t-sql/data-types/data-types-transact-sql?view=sql-server-ver15>

Following this, Normalize (3NF) the attributes into a number of tables.

Critique the rationale for the normalization strategy you produced.

II. Produce an Entity-Relationship model for this database, follow the style shown in <https://www.safaribooksonline.com/library/view/learning-mysql/0596008643/ch04s04.html>.

You may model these using MS Word, MySQL workshop, Lucidchart (online) or using Pen/Pencil and Paper

Hint: many to many relationships may be modelled with lookup tables which follow a generic form such as [relationshipID | keyIdentityA | keyIdentityB]. One to one relationships may be modelled with dedicated fields/attributes which link one tuple to another.

Task 2:

Using the azure environment, create the database tables you designed in Task 1.

Create simulated data to populate each table with at least one representative record.

Task 3:

What is the difference between the Serverless and Provisioned Compute tiers? Answer in your own words.

NOTE: A serverless instance offers a low cost deployment of an SQL server on Azure, alternatively a free tier instance exists which can be used with minimal cost concerns. In the below screenshot this is an S0 instance with 10 DTUs which stores up to 250gb.

This may change, ensure you verify free tier services @ <https://portal.azure.com/#blade/Microsoft Azure Billing/FreeServicesBlade> before instantiating resources, as this may vary with minimal notice.



NOTE: If Serverless is not selected then large costs may be incurred. Serverless consumes credit in a miniscule manner. Alternatively you may configure this within the free tier limitations (recommended).

At the time of writing this was S0, 10DTUs and 250gb storage. This is shown below.

Task 4 (Challenge Task):

Connect the Database using an application written in any programming language of your choice. You can get the startup for python using the following link.

<https://docs.microsoft.com/en-us/azure/azure-sql/database/connect-query-python>